

Lab - Shared Access Signatures (SAS) - At the Container Level

To Begin with the Lab

Summary

This lab demonstrates how to generate a **Container-level Shared Access Signature (SAS)** and use it in **Azure Storage Explorer**. A Container SAS provides secure, temporary, and permission-based access to all blobs within a specific container without exposing the storage account access keys.

Understanding

A **Container SAS** grants access to an entire blob container rather than a single blob. It allows you to specify permissions such as **Read, Write, Delete, List, Create, and Add**, along with a **start time** and **expiry time**. The SAS is signed using the storage account key (or a user delegation key) and can be used to securely connect tools like **Azure Storage Explorer**.

Unlike a **Blob SAS**, which provides access to only one file, a **Container SAS** allows users to work with multiple blobs inside the container according to the permissions assigned.

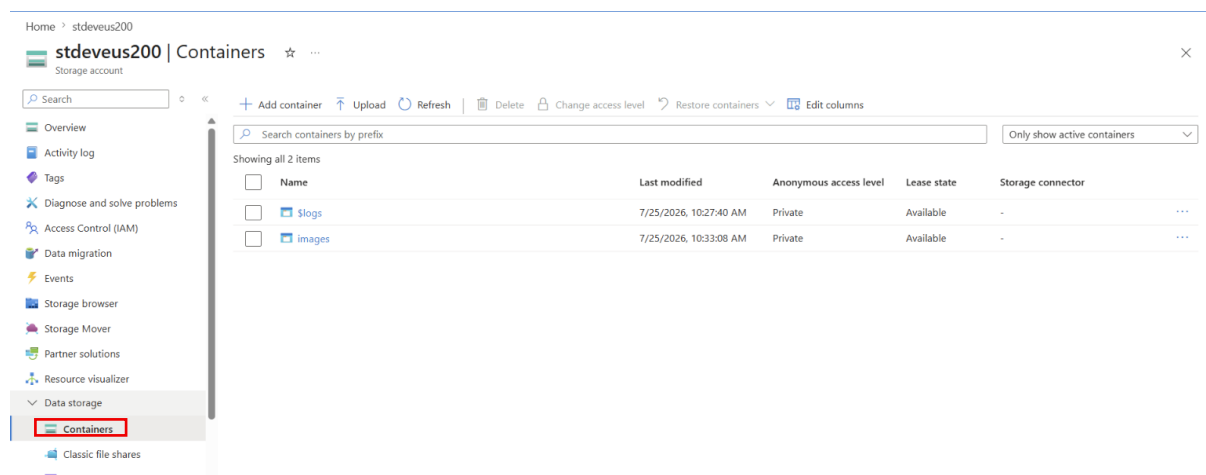
Example:

Suppose you want a team member to upload images to the **images** container for one week. Instead of sharing the storage account access key, you generate a **Container SAS** with **Read, Write, List, and Create** permissions that expires after seven days. The team member connects to the container using Azure Storage Explorer and can only perform the allowed operations during that time.

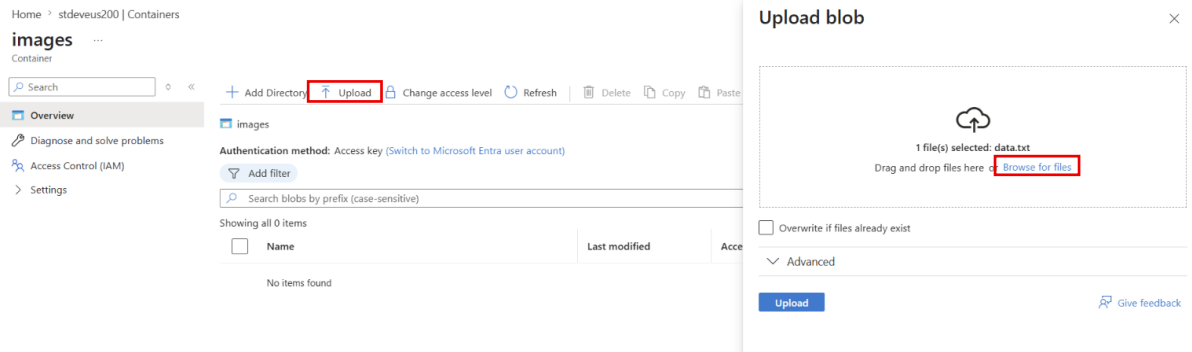
Lab Steps

Step 1: Open the Storage Account

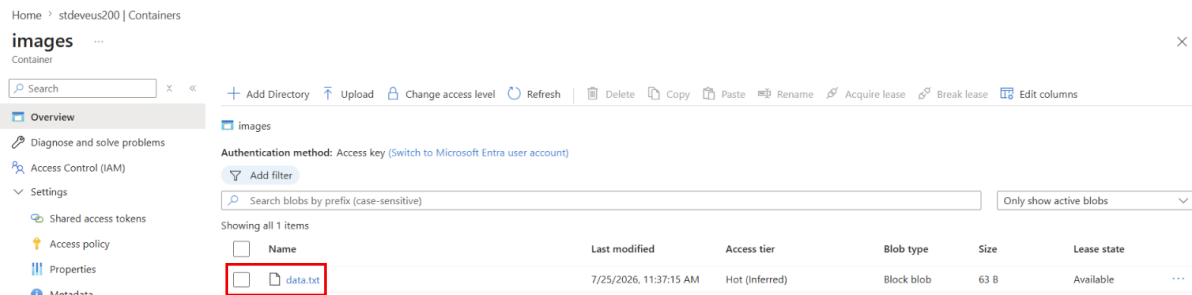
- Sign in to the Azure Portal.
- Open your **Storage Account**.
- Select **Containers**.



- Open the container(**images**).
- Click on the **Upload**.
- Click on the **Browse for File**.
- Choose **any File**.

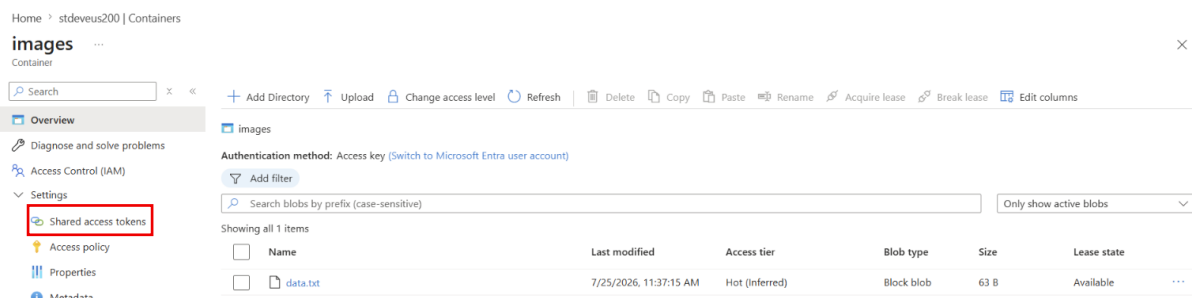


- **Uploaded File.**

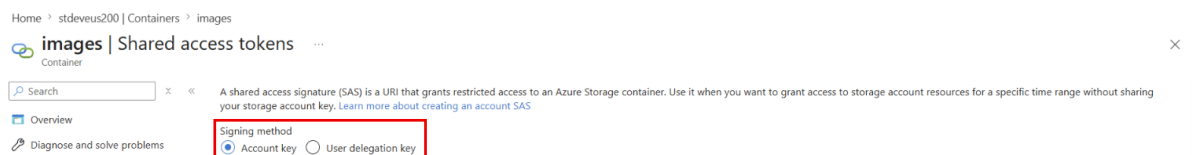


Step 2: Generate a Container SAS

- Select the container(**images**).
- Click on the **Shared access tokens** (left menu).



- Choose the **Signing Method as Account Key**.



Step 3: Configure Permissions

- Select the permissions required for the container.
- Common permissions include:
 - Read
 - List
- Grant only the permissions that are necessary.

A shared access signature (SAS) is a URI that grants restricted access to an Azure Storage container. Use it when you want to grant access to storage account resources for a specific time range without sharing your storage account key. [Learn more about creating an account SAS](#)

Signing method

☒ Account key ☐ User delegation key

Signing key ⓘ

Key 1 ▾

Stored access policy

None ▾

Permissions * ⓘ

2 selected ▾

☒ Read

☐ Add

☐ Create

☐ Write

☐ Delete

☐ Immutable storage

☒ List

Step 4: Set the Validity Period

- Specify the **Start Date and Time**.
- Specify the **Expiry Date and Time**.
- The SAS will only be valid during this period.

A shared access signature (SAS) is a URI that grants restricted access to an Azure Storage container. Use it when you want to grant access to storage account resources for a specific time range without sharing your storage account key. [Learn more about creating an account SAS](#)

Signing method

☒ Account key ☐ User delegation key

Signing key ⓘ

Key 1 ▾

Stored access policy

None ▾

Permissions * ⓘ

2 selected ▾

Start and expiry date/time ⓘ

Start
07/25/2026 11:36:48 AM

(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi ▾

Expiry
07/25/2026 12:00:48 PM

(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi ▾

Step 5: Generate the SAS URL

- Click **Generate SAS Token and URL**.
- Azure generates:
 - Container SAS Token
 - Container SAS URL

Signing method
☒ Account key ☐ User delegation key

Signing key ⓘ
 Key 1

Stored access policy
 None

Permissions * ⓘ
 2 selected

Start and expiry date/time ⓘ

Start
 07/25/2026 11:36:48 AM
 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi

Expiry
 07/25/2026 12:10:48 PM
 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi

Allowed IP addresses ⓘ
 for example, 168.1.5.65 or 168.1.5.65-168.1....

Allowed protocols ⓘ
☒ HTTPS only ☐ HTTPS and HTTP

Generate SAS token and URL

Step 6: Copy the Container SAS URL

- Copy the generated **Container SAS URL**.
- Keep it ready for use in Azure Storage Explorer.

Home > stdevius200 | Containers > images

images | Shared access tokens

Container

Search

Overview
 Diagnose and solve problems
 Access Control (IAM)
 Settings

Shared access tokens

Access policy
 Properties
 Metadata

Stored access policy
 None

Permissions * ⓘ
 2 selected

Start and expiry date/time ⓘ

Start
 07/25/2026 11:36:48 AM
 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi

Expiry
 07/25/2026 12:10:48 PM
 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi

Allowed IP addresses ⓘ
 for example, 168.1.5.65 or 168.1.5.65-168.1....

Allowed protocols ⓘ
☒ HTTPS only ☐ HTTPS and HTTP

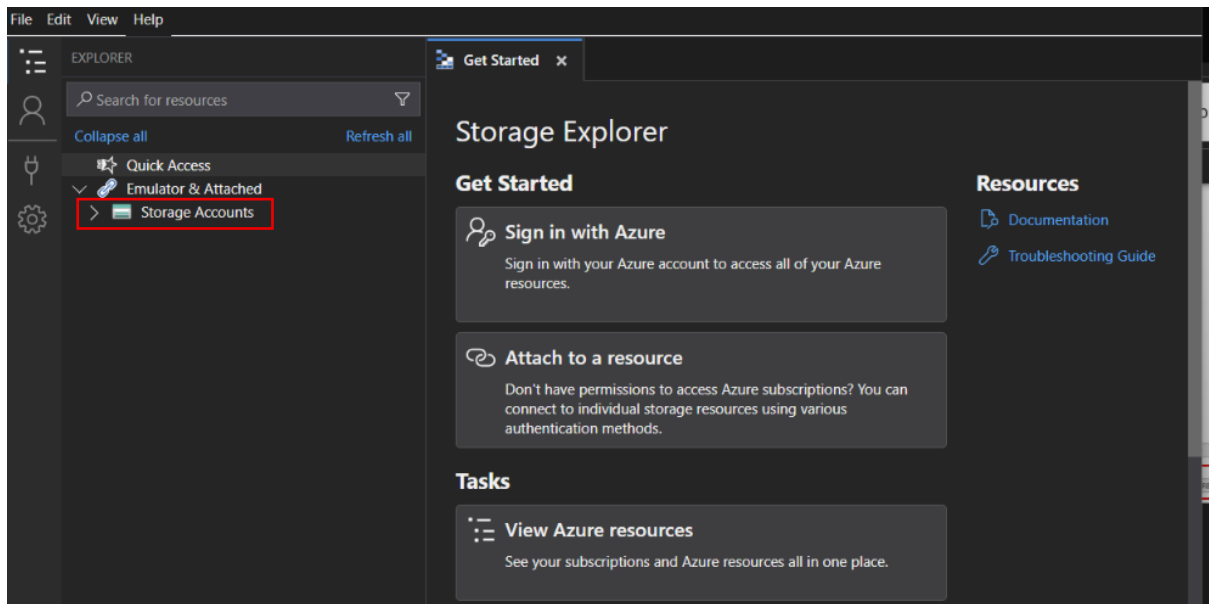
Generate SAS token and URL

Blob SAS token ⓘ
 sp=r&st=2026-07-25T06:06:48Z&se=2026-07-25T06:40:48Z&spr=https&sv=2026-02-06&sr=c&sig=oGwD7%2BjR8uChXICRytPnW8qgLxK6v3x%2BEcAuUa6TpU%3D

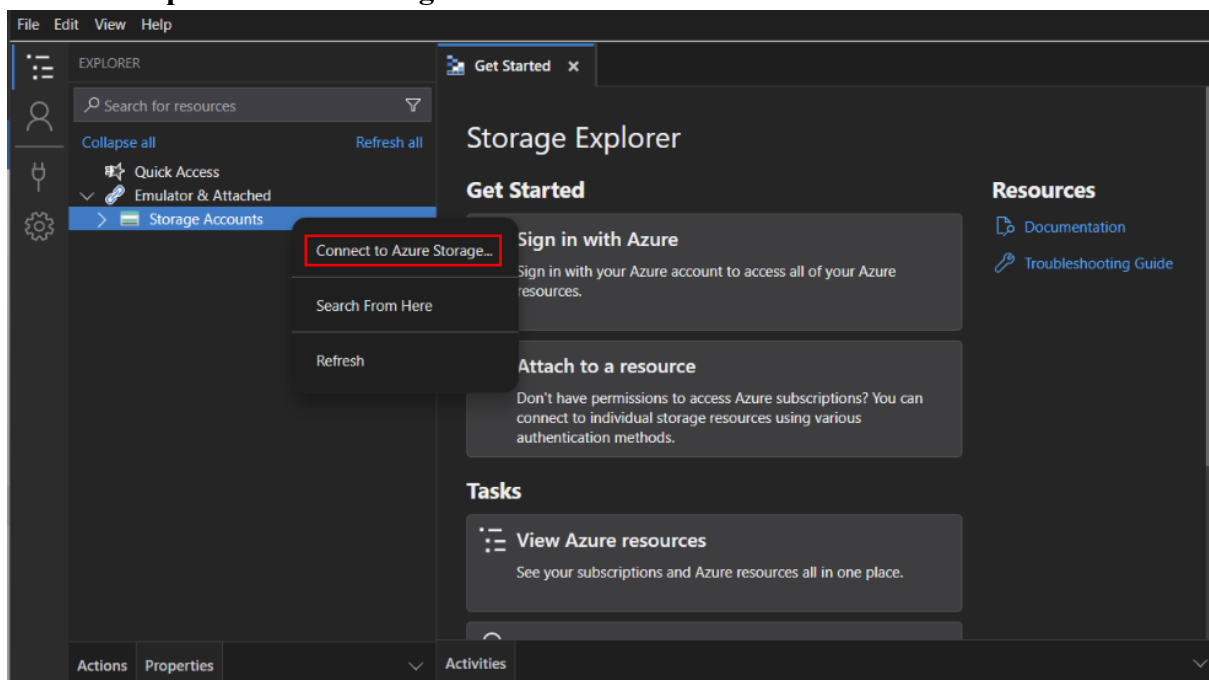
Blob SAS URL
<https://stdevius200.blob.core.windows.net/images?sp=r&st=2026-07-25T06:06:48Z&se=2026-07-25T06:40:48Z&spr=https&sv=2026-02-06&sr=c&sig=oGwD7%2BjR8uChXICRytPnW8qgLxK6v3x%2BEcAuUa6TpU%3D>

Step 7: Open Azure Storage Explorer

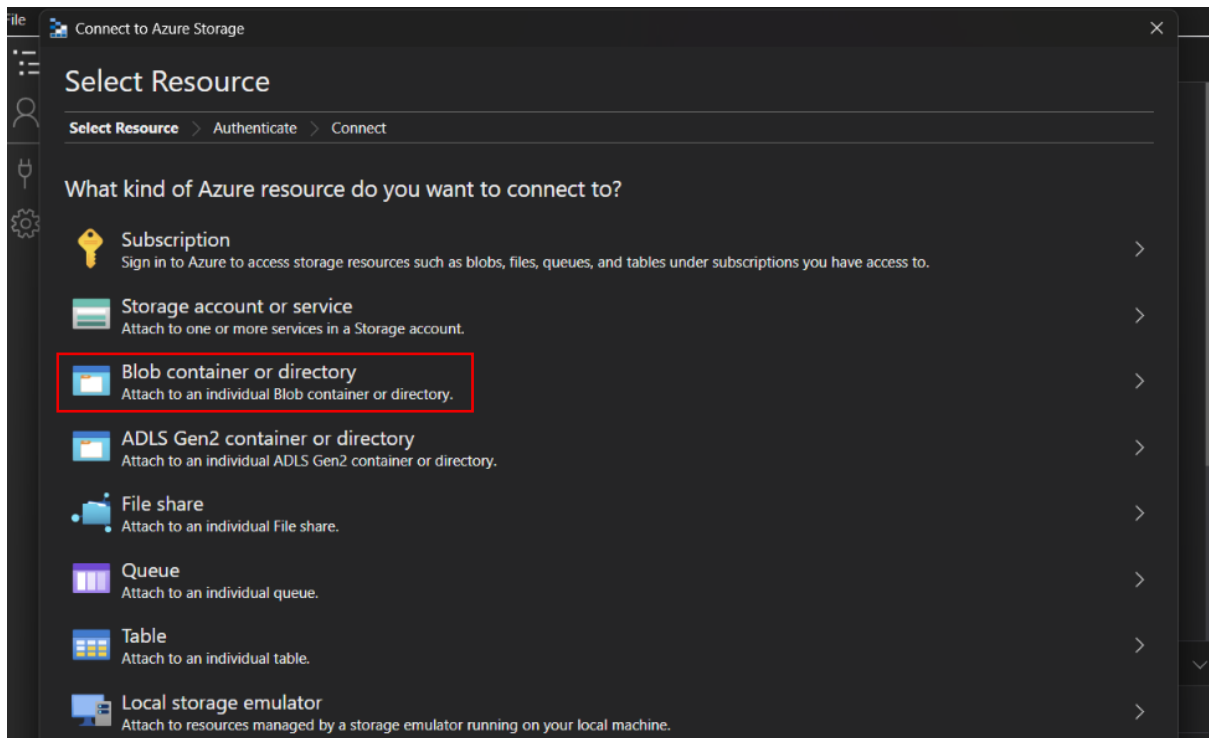
- Launch **Azure Storage Explorer**.
- Storage accounts(Right click).



- Select **Open Connect Dialog**.



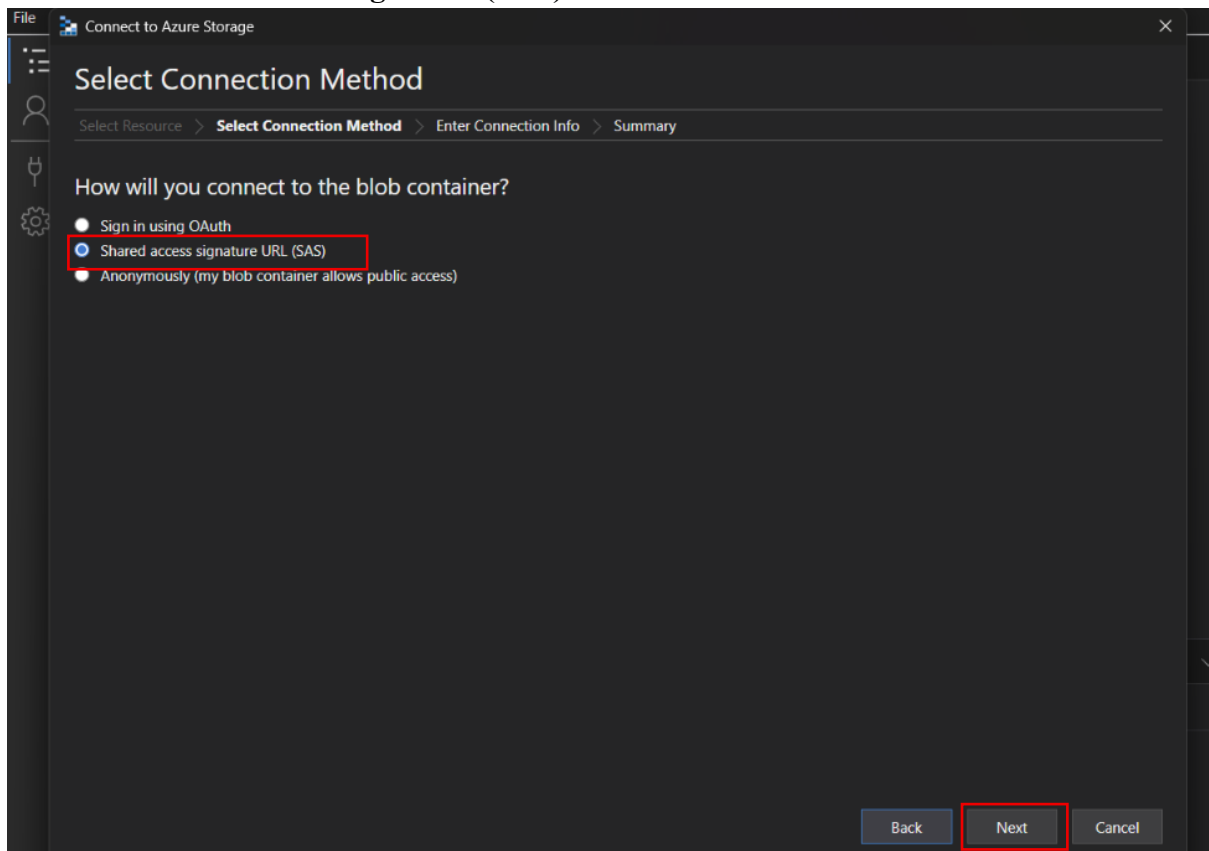
- Choose **Blob Container**.



- Click Next.

Step 8: Connect Using the SAS URL

- Select **Shared Access Signature (SAS) URI** as the connection method.



- Paste the copied **Container SAS URL**.

File Connect to Azure Storage

Enter Connection Info

Select Resource > Select Connection Method > **Enter Connection Info** > Summary

Display name:
images-1

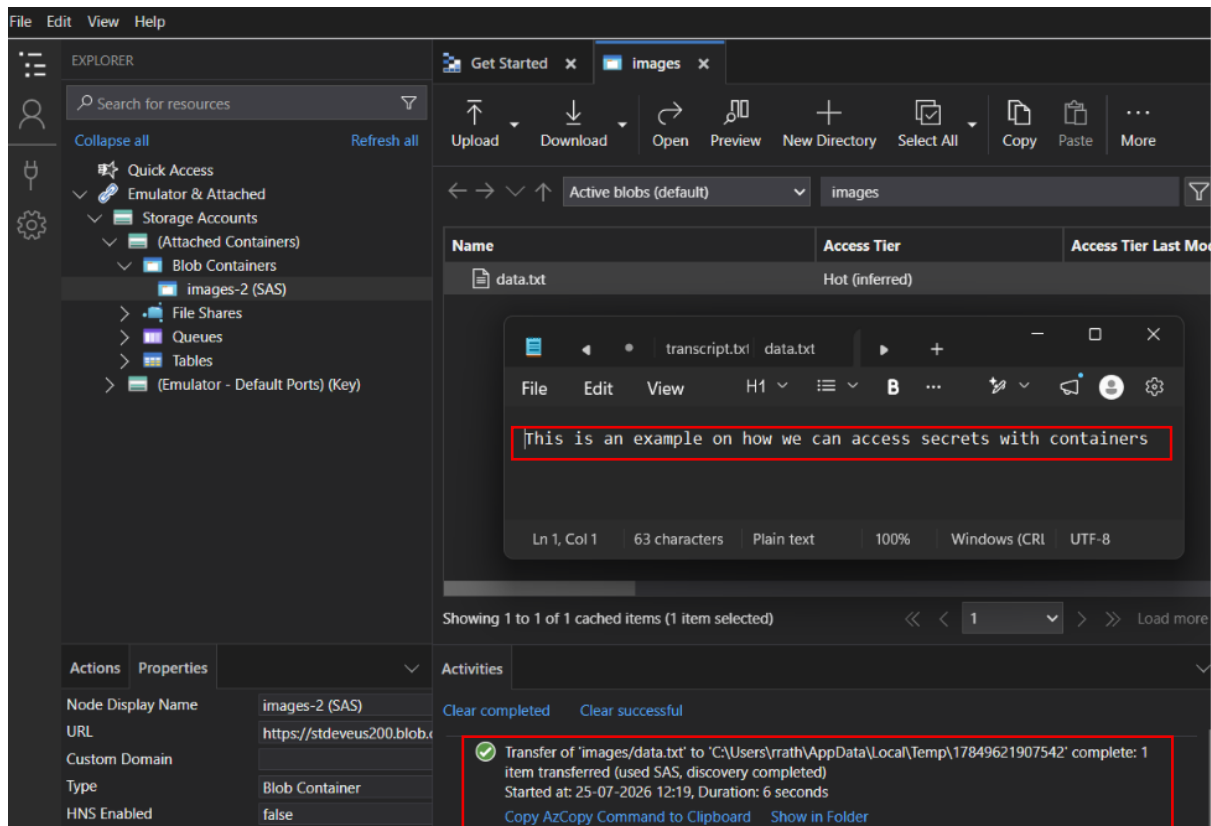
* Blob container or directory SAS URL:
`https://stdeveus200.blob.core.windows.net/images?sp=r&st=2026-07-25T06:06:48Z&se=2026-07-25T06:40:48Z&spr=https&sv=2026-02-06&sr=c&sig=oGwD7%2BfjR8uOhXICRytPnW8qgLxK6v3x%2BEcAuUa6TpU%3D`

Back Next Cancel

- Click **Next**.
- Review the connection details.
- Click **Connect**.

Step 9: Access the Container

- The container appears in Azure Storage Explorer.
- Browse the blobs inside the container.
- Perform only the operations allowed by the assigned permissions.
- Perform the **Read Permission**.



Step 10: Verify SAS Expiration

- Access the container before the expiry time.
- After the expiry time, try to reconnect or access the container.
- Azure denies access because the SAS has expired.

